University of Exeter

# MTH1001

## Mathematical Structures

| On Campus Exam | Closed Book |
|---|---|
| **Additional Materials:**<br>None | **Permitted Materials:**<br>Approved Non-Programmable Scientific Calculators Permitted (Green) |

**Guidelines:**

-The **duration** of this paper is 2 hours
-Word Count: No Word Count
-Write all answers in the answer booklet provided.
-Do not open the exam paper until instructed by the Invigilator.
-Use only black pen or pencil.
-Mobile and electronic devices are not allowed.
-Do not communicate with other candidates during the exam.

**Exam Instructions:**

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

## SECTION A

1. (a) For the permutations $\sigma = (135)(24)$ and $\tau = (12546)$ in $S_6$, find $\sigma\tau$, $(\sigma\tau)^2$ and $\sigma^2\tau^2$, expressing each as a product of disjoint cycles. Also, find the order and the sign of each of these permutations. (10)

   (b) Prove by induction that

$$\sum_{i=1}^{n} i(i+1) = \frac{1}{3}n(n+1)(n+2)$$

   for every natural number $n$. (10)

   (c) Let $n \in \mathbb{N}$ and $a, b \in \mathbb{Z}$.

   (i) What does it mean to say that $a \equiv b \pmod{n}$?

   (ii) Using your answer to (i), prove that if $a \equiv b \pmod{n}$ then $a^2 \equiv b^2 \pmod{n}$. (10)

   (d) Use the Extended Euclidean Algorithm to calculate $\gcd(599, 132)$ and to write it in the form $599a + 132b$ for suitable integers $a$ and $b$. Hence solve the congruence $132x \equiv 29 \pmod{599}$, or show that no solution exists. (10)

   (e) Let $G$ be a group (written multiplicatively).

   (i) What does it mean to say that a subset $H$ of $G$ is a *subgroup* of $G$?

   (ii) Let

$$H = \{h \in G : hg = gh \text{ for all } g \in G\}.$$

   Prove that $H$ is a subgroup of $G$. (10)

[**50**]

## SECTION B

2. (a) Give an example of a function $f : \mathbb{Z} \to \mathbb{Z}$ which is injective but not surjective. Justify your answer. (6)

   (b) Let $f : A \to B$, $g : B \to A$ and $h : A \to B$ be functions such that $g \circ f = \mathrm{id}_A$ and $h \circ g = \mathrm{id}_B$. Prove that $h = f$. (6)

   (c) List the elements of the group $\mathbb{Z}_{14}^{\times}$ of the invertible elements in $\mathbb{Z}_{14}$. Find the order and inverse of each element. Hence find all subgroups of $\mathbb{Z}_{14}^{\times}$. Explain how you know that there are no further subgroups of $\mathbb{Z}_{14}^{\times}$. (13)

   **[25]**

3. (a) Let $P$, $Q$, $R$ be propositions. Using a Truth Table, or otherwise, show that the propositions $((\neg P) \wedge Q) \Rightarrow R$ and $(\neg R) \Rightarrow P \vee (\neg Q)$ are logically equivalent. (6)

   (b) In each of the following cases, determine if the given function $\theta$ between groups is a homomorphism. Justify your answers.

   (i) $\theta : \mathbb{R}^{\times} \to \mathbb{R}^{\times}$, $\theta(x) = 2x$;

   (ii) $\theta : \mathbb{R} \to \mathbb{R}$, $\theta(x) = 2x$;

   (iii) $\theta : \mathbb{R}^{\times} \to \mathbb{R}$, $\theta(x) = x - 1$.

   Here, $\mathbb{R}^{\times}$ is the group of nonzero real numbers with the operation of multiplication, and $\mathbb{R}$ is the group of real numbers with the operation of addition. (9)

   (c) Let $p$ be a prime number with $p > 3$. Prove that $\sqrt{3p}$ is irrational. Any general results you use should be clearly stated, but need not be proved. (10)

   **[25]**

4. For any set $S$, we define an operation $\star$ on the subsets of $S$ by

$$A \star B = (A \cup B) \backslash (A \cap B).$$

So an element $s$ of $S$ is in $A \star B$ if and only if $s$ is an element of exactly one of $A$ and $B$.

(a) Let $S = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $C = \{1, 3, 5\}$. Write down each of the following sets as a list of elements enclosed in braces:

(i) $A \star B$;
(ii) $B \star C$;
(iii) $(A \star B) \star C$;
(iv) $A \star (B \star C)$. (6)

(b) Prove that the following properties hold for any set $S$, any subsets $A$, $B$, $C$ of $S$, and any element $s$ of $S$:

(i) $A \star \emptyset = A$;
(ii) $A \star A = \emptyset$;
(iii) $A \star B = B \star A$;
(iv) $s \in (A \star B) \star C$ if and only if either $s$ is in exactly one of the sets $A$, $B$, $C$, or $s \in A \cap B \cap C$;
(v) $(A \star B) \star C = A \star (B \star C)$. (14)

(c) Show that, for any set $S$, the powerset $\mathcal{P}(S)$ of $S$ is an abelian group with the operation $\star$. (5)

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**[END OF PAPER]**